

Tomas Pereira - AI Specialist

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WORK EXPERIENCE

Atlas Analytics Lab

May. 2023– Dec 2024

Master's/NSERC Researcher

Concordia University, Montreal, QC

- Work Under the Supervision of Prof. Mahdi S. Hosseini in the Field of Computational Pathology
- Research and Development of a Foundational Model for Breast and Colon Cancer Histopathology Tissue
- Research and Development of a Multimodal Vision-Language Model for Automatic Report Generation
- Recipient of the NSERC-USRA Research Award in May 2023

Concordia University

Sept. 2023 - Present

Teaching Assistant

Montreal, QC

- Teaching Assistant for the Machine Learning Course for both Graduate and Undergraduate Students

Bureau Veritas

July. 2022 – May. 2023

Systems Specialist

Montreal, QC

- Performed Data Wrangling and Data Cleaning Tasks to Assist Operation of Lab Analysts
- Refactoring and Maintenance of Datasheets and VBA Macros used in Daily Operations
- Key Figure in Development and Adoption of New Inventory System

EDUCATION

Concordia University

Sept. 2023 - Present

Master of Computer Science (MApCompSc)

Montreal, QC

- Recipient of the Concordia Merit Scholarship Award (2024 and 2025)
- Studied Representation Learning and Reinforcement Learning at MILA - Quebec AI Institute

Concordia University

Sept 2019 - May 2023

Bachelor of Computer Science (BCompSc)

Montreal, QC

- 2021 Recipient of the Government of Quebec Award under the Merit Scholarship Program

Projects

Research Project - Reinforcement Learning Multi Instance Learning For Efficient WSI Analysis

- Developed an Agent to Reduce the Computational Load of MIL Inference on Whole Slide Images Using RL

Research Project - Updating Method-Level Comments Using Generative AI

- Fine Tuned the CodeT5 and Gemma Models for Comment Generation and Updating.
- Studied the effectiveness of various inputs and gradual information gain

Research Project - Improving Performance of CNNs trained on Limited Data

- Studied the Effectiveness of Fine Tuning, Data Augmentation, and Synthetic Data on Small Datasets

Shipping Carrier Recommendation System

- Trained a Multiclass Classifier to be Used in a Logistics Company for Expediting Truck Selection. Led to Significant Reduction of Time Spent.

SKILLS & INTERESTS

Skills:

- o Languages: Python, Java, C++, VBA, Ruby, C#, SQL,
- o Technologies: Pytorch, SKLearn, Numpy, Pandas, SLURM
- o Other: Deep Learning, Neural Networks, R&D, Computer Vision, NLP, Decision-Making Systems, Medical Imaging, Data Science, LLMs
- o Spoken Languages: English, French and Portuguese (Trilingual)
- **Interests:** Dungeons and Dragons; Game Development; Gardening; Baking; Rock Climbing; Rugby;